Class: Algorithms & Data Structures Semester: Spring 2025

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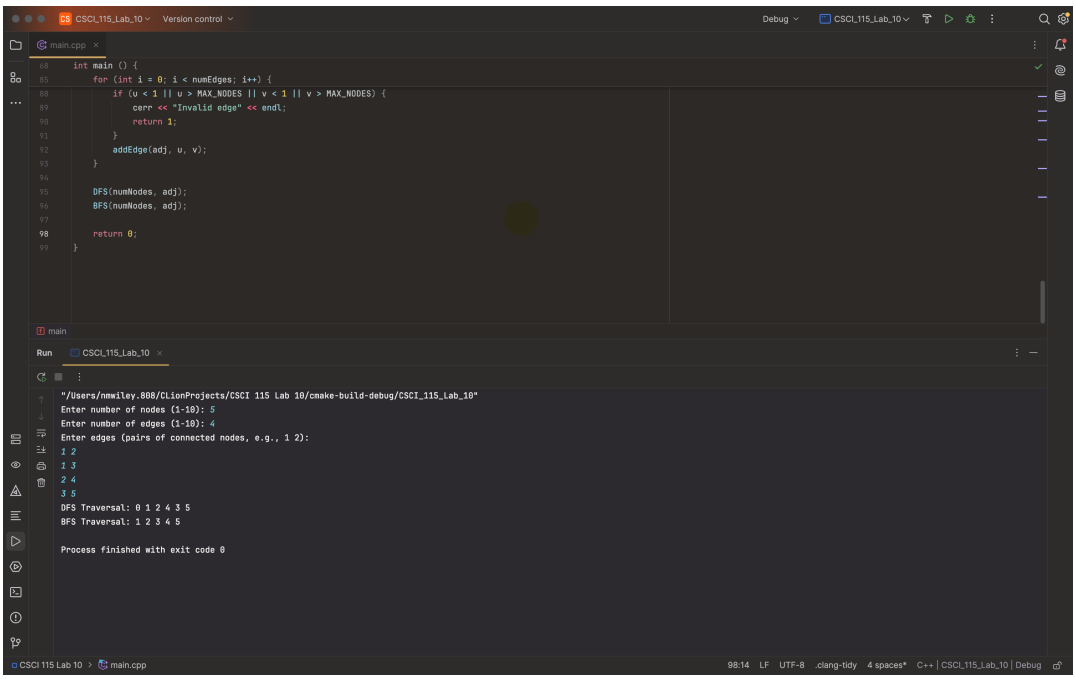
Lab Number - Section 09

CSCI 115 Lab 10 Report; BFS & DFS

- Statement Of Objectives
  - The objective of this experiment is to implement and understand the traversal mechanisms of a graph using Breadth-First Search(BFS) and Depth-First Search (DFS). The scope of this lab is limited to undirected graphs with a maximum of 10 nodes, labeled 1 through 10. The significance of this experiment lies in developing a deeper understanding of graph traversal techniques, recursive function design, and queue-based traversal. Major accomplishments include successfully implementing both DFS and BFS functions and displaying correct traversal order from a given set of inputs. This report details the experimental procedure, analysis, encountered challenges, and conclusions drawn from this experiment.

- Experimental Procedure
  - The experiment was approached by first defining the graph as an adjacency list using a vector of vectors. The graph is undirected, so the addEdge function adds connections in both directions. For DFS, a recursive approach was used, utilizing a boolean array to track visited nodes. For BFS, a queue was used to iteratively visit all neighbors of a node before proceeding deeper. The main function reads input from the user including number of nodes, number of edges, and each edge's connections.

- Analysis
  - The results of the experiment showed that both DFS and BFS traversals were completed accurately. For a sample input of 5 nodes and 4 edges (1-2, 1-3, 2-4, 3-5), the DFS traversal order was:
    - DFS Traversal: 0 1 2 4 3 5
  - And the BFS traversal order was:
    - BFS Traversal: 0 1 2 3 4 5
  - These outputs were consistent with expectations and demonstrates correct implementations of both algorithms



- Encountered Problems
  - One issue encountered was a linker error due to the incorrect use of const on the function signature of addEdge. The function tried to modify a const reference, causing the build to fail. This was resolved by removing the const qualifier. Other than this, the program compiled and executed correctly using the C++20 standard on macOS (CLion environment).

- Conclusions
  - Through this lab, I learned how to implement and apply BFS and DFS graph traversal algorithms. I gained a better understanding of how recursion and queue-based processing can be applied to graph structures. I also practiced debugging linker errors and managing input/output formatting for clean terminal display. This lab reinforced foundational concepts in graph theory and C++20 programming.

- References
  - Lab Slides - DFS & BFS